Executive Summary: Research Request Prioritization & ROI Scoring Tool

TL;DR

The Wake Forest School of Medicine (SOM) data management team currently receives a high volume of EHR data requests every day, prioritizing them by hand in a time-intensive process. This tool automates prioritization, scoring each request based on the principal investigator's research impact, funding status, and the team effort required—generating a defensible, data-driven ranked queue each day. The result: faster triage, smarter allocation of resources, and a measurable increase in support for high-impact research.

Goals

Business Goals

- Direct limited data management team capacity toward the highest-impact medical research at Wake Forest.
- Establish a defensible, auditable methodology for request prioritization that can be communicated to leadership and stakeholders.
- Identify niche and high-growth medical research specialties where Wake Forest can develop strategic advantage.
- Build an institutional dataset that improves over time via machine learning.

User Goals

- Free data managers from manual research so they can focus on high-value request fulfillment.
- Give the team lead simple, configurable control over prioritization criteria—no technical skills required.
- Ensure users act on clear priorities, reducing confusion and delay.
- Provide transparent feedback on how prioritization is determined.

Non-Goals

- The tool does not make final or irrevocable decisions; it empowers and informs human judgment.
- No patient-level data is used or exposed in any workflow.
- The tool does not automate fulfillment or project management—only triage and prioritization.

User Stories

Data Manager

- As a Data Manager, I want to arrive each morning to a ranked queue of requests, so I can immediately focus on the highest-impact work without spending hours on manual PI research.
- As a Data Manager, I want to export the prioritized queue for meetings, so my team can act efficiently.

Team Lead

- As a Team Lead, I want to adjust scoring priorities (e.g., weigh funding more during grant season) and see the queue update in real time, so our process always reflects current strategy.
- As a Team Lead, I want to save and reuse common scoring configurations, so I can easily match our approach to evolving needs.

Leadership/C-Suite

- As a Leader, I want evidence that our data management capacity is directed toward research most likely to produce publications, grant renewals, and enhance Wake Forest's reputation.
- As a Leader, I want reporting that shows which specialties and research areas are most served, enabling better strategic investment.

Functional Requirements

- **Automated Request Scoring** (Priority: High)
    - Pulls public information about each PI (NIH grants, citation metrics, journal prestige) and generates a composite ROI score—no manual research needed.
- **Effort-Adjusted Prioritization** (Priority: High)
    - Scores account for both estimated return (PI research impact) and investment (team hours required), delivering a true ROI ranking.
- **Team Lead Controls** (Priority: High)
    - Simple slider controls allow adjustment of weighting (e.g., more grant emphasis during renewal season); queue dynamically updates. Named configurations can be saved.
- **PI Research Profile** (Priority: Medium)
    - Each PI gets a continuously updated dossier: publication history, grant funding, journal impact, request completion, and publication rate from previous requests.
- **Strategic Market Intelligence** (Priority: Medium)
    - Each project's research area is classified by medical specialty and contextualized with national market size, revealing strategic value areas.
- **Historical and New Request Coverage** (Priority: High)
    - All historical requests are scored at launch. New requests are scored immediately on upload. Brand new PIs get estimated scores from faculty rank and funding source until more data is available.

- **Simple Export for Meetings** (Priority: Medium)
    - One-click Excel export of the open ranked queue for team and leadership reporting.

User Experience

Entry Point & First-Time User Experience

- Users access the tool via a secure institutional web app.
- Onboarding is intuitive: upload the existing weekly REDCap request tracking file; an initial tutorial highlights scoring and export features.

Core Experience

- **Step 1:** The Data Manager uploads the weekly REDCap export.
    - Simple interface; minimal friction.
    - Error handling checks for file type/format compatibility.
    - Flags success/failure; suggests corrections if needed.
- **Step 2:** The tool automatically retrieves all necessary PI data, computes ROI scores, and ranks requests in the background.
    - All processing is invisible to users; status bar shows progress.
    - Users notified when the ranked queue is ready.
- **Step 3:** The Data Manager reviews the ranked open request queue.
    - Clear display: project title, PI, funding amount, specialty, score breakdown.
    - Export to Excel in one click.
    - Team Lead can open the configuration panel, adjust sliders, and see rankings update instantly.
    - Both users can override rankings if needed; overrides are tracked for transparency.

Advanced Features & Edge Cases

- New PIs or requests without historical data receive estimated scores based on faculty rank and funding source, clearly labeled in the interface.
- Configuration changes are auditable; prior versions of ranking logic can be revisited for transparency.
- Advanced users can review detailed PI research profiles for context when making override decisions.

UI/UX Highlights

- Clear visual differentiation between estimated, fully-enriched, and manually-overridden scores.
- Responsive, accessible design for all device types.
- Color-coding emphasizes action items and confidence levels.
- Quick help icons provide definitions for scoring fields and specialties.

Narrative

Wake Forest School of Medicine’s data management services are critical in enabling groundbreaking medical research. Yet without a systematic approach to prioritizing EHR data requests, the team’s finite capacity is allocated inconsistently—sometimes dedicating resources to low-impact projects while studies with major grant funding or publication potential wait.

This new prioritization tool delivers a step change: every day, each incoming research request is scored and ranked based on public, objective criteria—NIH funding record, publication impact, and anticipated effort. The result is a transparent, justifiable queue where the highest-potential projects naturally rise to the top. Managers control strategic weighting through a straightforward interface, instantly aligning team priorities with institutional goals without requiring technical knowledge.

Over time, leadership gets more than just anecdotal assurance: they enjoy clear visibility into which requests, specialties, and researchers are being served—and can see in hard numbers where Wake Forest is developing competitive strength. As the institutional dataset grows, the system will soon be able to forecast the likely impact of a project the moment a request arrives, enabling true strategic resource management. What was once ad hoc, labor-intensive triage evolves into a forward-looking, data-driven engine for institutional research excellence.

Success Metrics

User-Centric Metrics

- 50% reduction in manual triage time within 3 months of launch (self-reported and time-logged).
- 80% of data managers using the ranked queue as their main workflow tool within 3 months.
- Team satisfaction score of 4/5 or above in post-launch surveys.

Business Metrics

- Measurable increase in proportion of completed requests benefiting federally-funded or high-impact studies (tracked quarterly).
- Increase in publication rate from requests serviced (compare 12-month post-launch data to previous baseline).
- Decrease in escalations or complaints about prioritization decisions.

Strategic Metrics

- Identification of niche or high-opportunity specialties for strategic planning.
- Institutional dataset of PI histories and request outcomes built for ML model training in 12-18 months.

Technical Metrics

- 99.9% uptime on scoring and ranking features during business hours.
- API success rate >99%.
- Export files delivered in under 10 seconds for up to 500 open requests.

Tracking Plan

- Number of uploads/exports per week.
- Frequency and rationale for manual overrides.
- Distribution and evolution of ROI scores across requests.
- Team Lead adjustments to scoring weights.
- Adoption rates among data managers.

Technical Considerations

Technical Needs

- Integration with REDCap export and Trac hours log (existing team data).
- API calls to NIH RePORTER, PubMed, Clarivate JCR, CrossRef, and Google Scholar Metrics.
- Cloud-based or on-premise server to process, store, and retrieve PI scoring data.
- Front-end dashboard for queue triage, configuration, and export.

Integration Points

- REDCap (Excel export).
- Trac (team time logs).
- Wake Forest's Clarivate JCR institutional subscription.

Data Storage & Privacy

- Stores only faculty (PI) profiles, project metadata, and publication/grant info.
- No patient- or subject-level health data stored.
- Compliant with internal Wake Forest policies and NIH/data source guidelines.
- Pending final data retention policy review.

Scalability & Performance

- Designed for daily queue processing of 100–500 active requests.
- Asynchronous processing and caching on all enrichment calls.
- Scalable to accommodate higher load as request volume grows.

Potential Challenges

- Occasional missing data for new PIs (addressed with proxy scoring and confidence discount).
- Institutional Clarivate API limits (monitored as part of launch QA).

- Timely updates from external APIs; robust error handling in case of service disruptions.

Milestones & Sequencing

Project Estimate

- **Medium:** 3–4 weeks to functional prototype.

Team Size & Composition

- **Small Team:** 1 backend developer, 1 frontend developer, 1 product stakeholder (Data Manager lead).

Suggested Phases

Phase 1: Core Data Pipeline (Weeks 1–2)

- Key Deliverables: Backend developer builds automated PI research data retrieval, initial scoring engine, and ranked queue output.
- Dependencies: Sample REDCap/Trac exports, API key setup.

Phase 2: Dashboard and User Interface (Weeks 2–3)

- Key Deliverables: Frontend developer and product stakeholder build dashboard display, score breakdowns, simple configuration panel, and override logic.
- Dependencies: Core scoring engine API from Phase 1.

Phase 3: Full Integration & Launch Readiness (Weeks 3–4)

- Key Deliverables: End-to-end integration, robust error handling, audit logging, Excel export. Historical request backfill completed.
- Dependencies: All prior phases completed.

V2 Enhancement (12–18 months post-launch)

- ML model trained on accumulated outcomes to predict expected ROI of new requests automatically, enabling fully proactive triage.
